

Quantum Toom's Contours

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Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math

Our Code

Code

Consider the representation of the d -dimensional toric by the cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ ¹.

In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

¹where 1_i is the vector consisting of one at the i th coordinate and zero elsewhere

Local Decoder

Positive edge

We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1 \neq g_2 \in S$, if $x = v$ and y is either $g_1 v$ or $g_2 v$.

Toom's rule

We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.

Let $\sigma^{(t)}$ be a domain wall at time t , and let $\Gamma^t = \{v : \exists \{v, u\} \in \sigma^t\}$. We denote by ϕ^t the subsets of bits which are flipped as a result of applying the Toom's rule at time t , and by ϕ_v the bits which are flipped by vertex v . We say that u is added to Γ^{t+1} by v at time t , if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v)$. When it clear from the context, we will omit the time, and say that u is added by v .